

Berlin vs. London Airbnb Market Analysis

Data Visualization Final Project

Batul Elyas Lokhandwala

[GitHub Repo](#)

[Live Streamlit App](#)



Made with GAMMA

What We Set Out to Understand

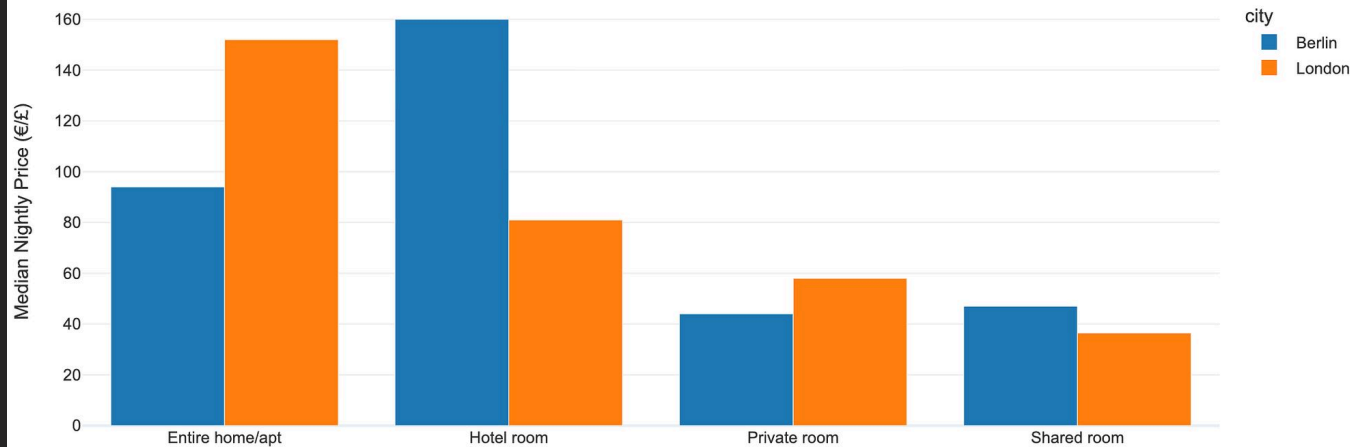


This project compares Airbnb listing data across **Berlin** and **London** — two of Europe's most dynamic short-term rental markets. Using public listing data, we explore pricing dynamics, host behavior, stay duration patterns, and geographic distribution.

📄 **Core Finding:** London commands higher median prices and greater professional host density, while Berlin features more single-listing residential hosts — reflecting distinct market structures.

Pricing Across Room Types

London Commands Higher Median Nightly Rates Across All Accommodation Types



Insight: London listings consistently command higher nightly rates across all categories compared to Berlin, with Entire Homes showing the largest price premium.

A boxplot of listing prices segmented by room type reveals clear stratification between the two cities.

Entire Homes Lead

Entire home/apartment listings carry the highest median prices in both cities, with London significantly outpacing Berlin.

Shared Rooms: Minimal Spread

Shared rooms show the tightest price distribution — low variance, low median — confirming their budget positioning.

London Premium

Across all room types, London listings command a noticeable price premium, reflecting higher demand and operating costs.

Commercial Host Density: Who Owns the Listings?

London Market Features a Higher Density of Multi-Listing Professional Hosts



Insight: London shows a significantly wider upper distribution of host listings, indicating a higher presence of commercial operators and multi-listing managers compared to Berlin's more localized single-host baseline.

A violin plot of listings per host exposes the professional vs. residential host divide in each market.

London: Professional Market

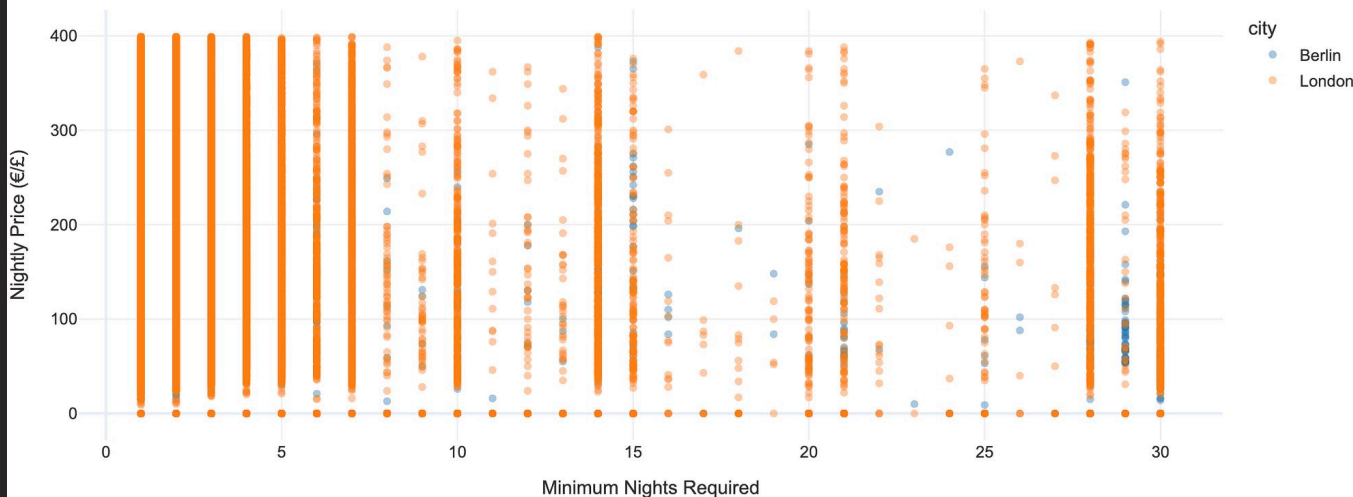
A significant tail of hosts managing 10+ listings points to commercial operators and property management firms.

Berlin: Residential Market

The distribution is heavily concentrated at 1–2 listings per host, indicating individual homeowners and casual hosts.

Stay Duration vs. Pricing Strategy

Short-Term Stays (1–3 Nights) Dominate Upper Price Tiers in Both Cities



Insight: Premium pricing is heavily concentrated among listings with short minimum stay requirements (1–3 nights), whereas listings requiring 30+ nights consistently cluster in lower rate bands.

→ Short Stays = Higher Nightly Rates

Listings with low minimum-night requirements (1–2 nights) tend to charge higher per-night prices, targeting tourists.

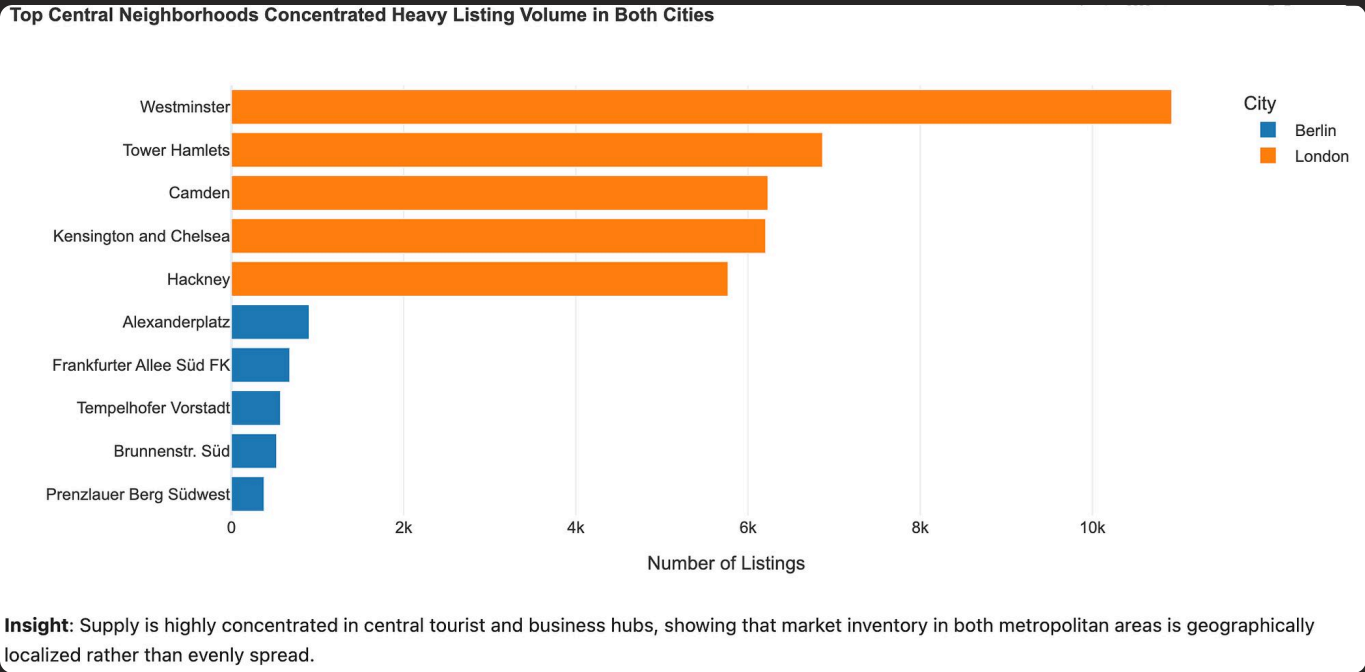
→ Longer Stays Discount the Rate

Hosts requiring 7+ night minimums typically offer lower nightly rates, incentivizing extended bookings.

→ City Contrast

London shows a steeper price drop-off for long stays than Berlin, suggesting stronger tourist-driven pricing.

Where Are the Listings? Geographic Distribution



A bar chart of top neighborhoods by listing count reveals where Airbnb activity concentrates in each city.

London: Central Clustering

Listings concentrate in central boroughs — Westminster, Camden, and Kensington — reflecting tourist demand hotspots.

Berlin: Distributed Pattern

Berlin's listings spread across Friedrichshain, Mitte, and Kreuzberg, showing a more even urban distribution.

Implication

London's geographic concentration may amplify housing pressure in specific neighborhoods compared to Berlin.

DASHBOARD SHOWCASE

Interactive Streamlit Dashboard

The live dashboard brings all four analytical questions to life with dynamic filtering and real-time exploration.



City Selection Filter

Toggle between Berlin and London instantly to compare metrics side-by-side.



Dynamic Metrics

Key statistics — median price, listing count, host density — update based on filter selections.



Tab Navigation

Organized tabs separate each analytical question for a clean, guided exploration experience.

Filter Options

Select Cities:

Berlin x London x



Airbnb Market Comparison: Berlin vs. London

An interactive analysis exploring price distributions, commercial host density, and listing metrics across two major European hubs.

Share ☆ ↻ ⋮

Total Listings Analyzed

105,654

Overall Median Price

€/£ 173.00

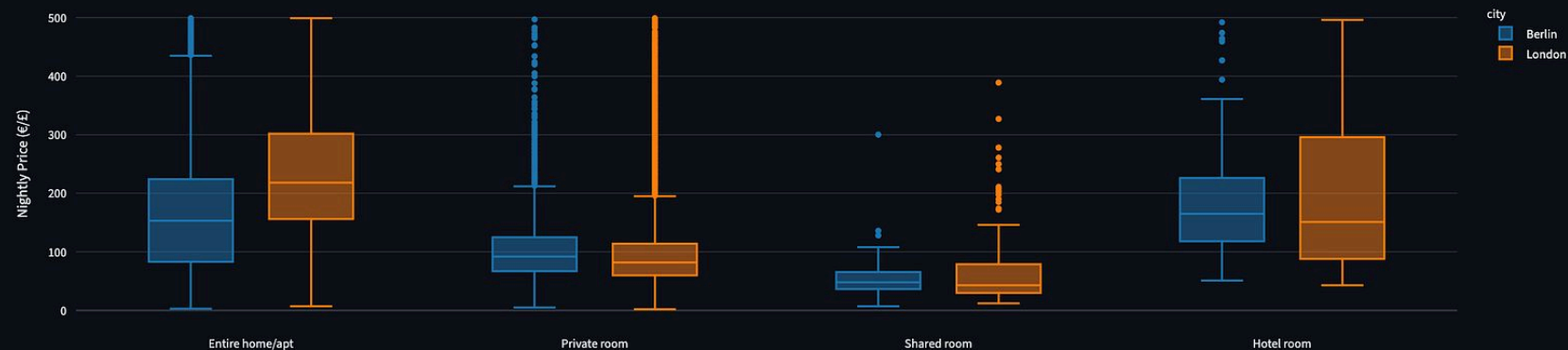
Avg Reviews / Month

1.04

💰 Pricing & Host Density 📅 Stay & Availability 📍 Geography

Figure 1: Cross-City Price Distribution by Room Type

London Commands Higher Median Nightly Rates Across All Room Types



Key Findings & Takeaways



London Premium

Higher median prices across all room types reflect London's stronger tourist demand and cost pressures.



Professional Hosts

London's long tail of multi-listing hosts signals a more commercialized short-term rental market.



Berlin Residential

Berlin's host base is predominantly single-listing homeowners, suggesting a more community-rooted market.



Geographic Contrast

London's centralized clustering vs. Berlin's distributed pattern reveals different urban housing dynamics.